Round 9
Architecture assessment

Fixed-weight ReLU MLP · 1024 × 16 · Research response · 23 September 2026

One useful redesign; no demonstrated solution to both targets

Conclusion. I cannot derive an estimator that achieves both 133-165 u and raw MSE 1.6-1.9×10⁻⁸ from the supplied evidence. I can derive a different split of the cubic/quartic work: **keep the third leg of the cubic readout in the previous coordinate system**. It removes the two extra per-source projections for a $K_3 \times C$ response. It does not halve the 260 u cubic chain, and a complete ledger does not yet support the requested ≤20 u quartic increment.

Labels: Derived denotes algebra or arithmetic; **Conjecture** denotes unverified network benefit; **Measure** denotes a falsifier. Your Round 9 measurements are accepted as supplied. In particular, teacher-forcing now closes the bivariate-closure question; proposing another covariance-closure patch would ignore that result.

1. Mixed-coordinate response architecture

Derived. Use row-oriented weights B, so the next pre-activation is Bh and its i-th row is w_i^T. At the current ReLU let $D = \mathrm{diag}(d_1)$, $d_1 = \Phi(\mu/\sigma)$, and let K be the current pre-activation third cumulant. Separate its linearly transmitted part $\bar{K} = D \otimes^3 K$ from the newly born post-ReLU source. Carry the transient mixed slice

$$R_{ik}^{\mathrm{old}} = \bar{K}(w_i, w_i, e_k), \qquad D21_{\mathrm{linear}} = R^{\mathrm{old}} B^T.$$

For a normalized symmetric source Sym $\Sigma_r(p_r \otimes p_r \otimes a_r)$, denote its pre-B, gain-scaled legs by P,Ā and its transported legs by $P' = BP$, $A' = B\bar{A}$. Then

$$R_s = \tfrac{1}{3}\left[(P' \circ P')\bar{A}^T + 2(P' \circ A')\bar{P}^T\right].$$

The source's usual D21 is exactly $R_s B^T$. Sum R_s before applying this final B^T. The actual V29 source includes AAP, MPP and thin terms: apply the same rule to every third right leg; do not replace its source by the illustrative PPA formula. The left Hadamard factors are unchanged.

State. Keep the existing mean, covariance, K_3 sources and quartic-kernel inputs, plus R_old and R_total (two n^2 float32 arrays: 8 MiB). Retain pre-B right legs and the pre-B shared basis until readout finishes. With four young sources, copied pairs cost 32 MiB; basis and response scratch bring the incremental working set to roughly 50-70 MiB. Buffer reuse can reduce this; measure actual peak memory.

Why the realized fluctuation survives. This is a change of contraction order, with the same legs and weights. It neither averages the odd contribution nor compresses the Hadamard products. Under $B \to -B$, R is even and RB^T is odd. Every retained all-distinct cubic entry is preserved, including its existing approximation error.

2. Propagation and the quartic response readout

Derived: shared old basis. If pre-B legs are Q_-F and post-B legs are BQ_-F, form the usual source-summed inner contraction S, but assemble $R = SQ_-^{\top}$ instead of $D21 = S(BQ_-)^{\top}$. The n^2r work is the same. Preserve Q_- *after* the gain/refit and its nested sub-basis; otherwise this silently changes the retained tensor. No inverse of B is required.

Derived: newborn cancellation. The newborn has $P_-=I$. Its mixed readout is $L_A\, a_b^{\top} + L_P$, whereas its ordinary readout uses two dense right-factor products. Saving that one newborn product pays for the single aggregate $R_{total}\, B^{\top}$ product. Thus, on layers already computing full D21, this change has no additional square-matmul bill. Thin terms must be remapped too.

The public source forms a_b = D C_off [1]. Its already transported newborn A leg therefore supplies the covariance response, including an elementwise diagonal correction:

$$q_i = CDw_i, \qquad q_{ik} = (Ba_b)_{ik} + B_{ik} d_{1,k} C_{kk}.$$

Derived: $K_3 \times C$ readout. Set $u_i = Dw_i$, $d_2 = \varphi(\mu/\sigma)/\sigma$, $t_i = (d_2 \circ w_i) \circ q_i$. The leading connected contribution containing one quadratic response, one K_3 and one covariance is

$$\Delta\kappa_{4,i}^{(3+2)} = 12K(t_i, u_i, u_i) = 12\sum_k R_{ik}^{\mathrm{old}} v_{ik},$$

$$v_i = (d_2/d_1) \circ w_i \circ q_i.$$

The factor 12 is 4 choices of the quadratic output × its Taylor factor 1/2 × 6 connected K_3-C partitions. Evaluate d_2/d_1 with a stable inverse-Mills ratio, not an underflowed Φ division. R_old must exclude the newborn: this response is driven by the *input* K_3. This single rowwise contraction replaces projecting two source legs along every new curvature direction.

Derived: Gaussian pieces. With $c_i = d_3 \circ w_i$ and $d_3 = -(\mu/\sigma)\varphi(\mu/\sigma)/\sigma^2$, the leading star and path readouts are

$$\Delta\kappa_{4,i}^{G} = 4\sum_k c_{ik} q_{ik}^3 + 12 t_i^T C t_i.$$

The star is elementwise; evaluating Ct for all queries costs one square product per transition. These formulas specify connected polynomial terms, **not the entire quartic estimator**. Preserve the input-K_4 edge, pair-sector matching and conventions of your measured −8.8% kernel. Their costs and any subtractions cannot be declared free.

Conjecture, narrowly stated. If this readout reproduces that same measured kernel on the same retained tensor, it inherits its κ_4 improvement. It does not improve K_3 all-distinct birth accuracy or solve the unexplained 55%. No reduction of fresh error in all three channels is established by this architecture.

3. Complete cost boundary: where “free” stops

Derived. All costs below are incremental to 260 u. A square product is 0.51 u with the supplied Strassen implementation. A classical (n,n)×(n,r) product costs $n\,r(2n-1)/2^{31} \approx r/n$ u. Hadamard/reduction work is counted separately; no $n\times r^2\times n$ product is relabeled n^2r.

Operation	Per applicable layer	Whole-network contribution
Aggregate remap RB^T	+0.51 u	Canceled by next row in full-D21 layers
Newborn identity right leg	−0.51 u	One fewer newborn dense contraction
q reuse; full K_3 response	$O(n^2)$	About 0.01-0.02 u/layer for basic readout algebra; meter tails/thin terms
Gaussian path Ct	0.51 u	≤7.65 u for 15 transitions
Literal all-distinct input-K_3 projection, if required	1.02 u	≤15.30 u; two products unless demonstrably reused
Restore final-layer mixed slice	Terminal only	Roughly 9-11 u, plus thin/alignment costs
Input-K_4 edge; output pair matching	Kernel dependent	Retain and meter reference operations; not included as zero

Derived: repeated indices. If “all-distinct” means retaining only distinct indices of the input $\mathcal{K}$, write $J_{jk}=\mathcal{K}_{jjk}$ and $j_k=\mathcal{K}_{kkk}$. The exact subtraction is

$$R_{\text{ad}} = R - (B \circ B)J - 2B \circ (BJ^{I}) + 2(B \circ B) \circ j^{I}.$$

It costs two dense products. These input-index exclusions are different from matching repeated *output* indices of K_4. If your 140 u implementation uses the latter, audit its actual formula; the simple row dot above cannot inherit the −8.8% claim until it matches that formula.

Derived: terminal tax. V29 calls the final readout with need_d21=False [1]; it computes D3 without dense slice contractions. R_old would restore them. An illustrative bill is 1.02s_dense + 2Σ_old(r_s/n) + r_common/n u, plus nested-basis lifts and thin terms. Four dense sources and the stated rank tiers give roughly 9-11 u. Count only eligible input sources and use the actual terminal inventory.

With literal all-distinct subtraction, the visible increment is approximately 7.65+15.30+(9-11) = **32-34 u before remaining overhead**. Omitting that subtraction yields roughly 17-19 u before overhead, but that is a different kernel unless equality is proved. I therefore do **not** certify ≤20 u.

Score consequence. If raw MSE falls by 8.8%, the score ratio is 0.912(1+c/260), where c is added cost. Break-even is c=25.09 u. At c=20, score improves only 1.8%; at c=35, it worsens 3.5%. Raw MSE would be 1.915×10^{-8}. This does not approach the leaders’ cost.

4. A falsifier before any architecture claim

Measure: one stored late layer. Capture layer 14's source legs immediately before and after raw B, post-gain/refit bases, newborn a_b, left Hadamard factors, thin legs and the direct quartic-kernel output. The listed marginal/slice arrays alone do not identify the all-distinct tensor; source capture is required and is allowed by your setup.

First compare R_total B^T with the existing full D21, including its diagonal before any masking. Compare the rowwise K_3 response with the direct per-source curvature contractions, then compare the *complete* quartic prediction with the measured 140 u version, including exclusions and pair matching. A relative RMS discrepancy above 10^{-5} is a provisional failure; check float32 accumulation order against a small float64 reference before diagnosing algebra.

Second run the actual final-layer branch and the full metered operation graph. Reject the score-improvement claim if added cost exceeds 25.1 u for the same −8.8% raw gain. One-layer replays and one 16-layer run need no new Monte Carlo; they are an under-hour CPU falsifier on hardware capable of the stated baseline. Separately enforce the grader's 120 s and 8 GB limits.

End-to-end test. Keep all coefficients fixed. The rewritten estimator must reproduce the direct augmented estimator's moment trajectories and final means to numerical tolerance before the deterministic paired 100-network test. A match establishes a computational reorganization, not improved K_3 modeling. Any subsequent joint K_3/K_4 redesign must be scored independently; do not add channel-oracle percentages.

Why I do not offer two more "new" architectures

Derived: explicit all-distinct birth. A packed symmetric K_3 has 179,481,600 float32 entries, about 0.718 GB. The direct three-mode transport algorithm costs ≈3n u = 3072 u per layer classically, or ≈1567 u using tiled 0.51 u square products, before nonlinear birth. Symmetry helps constants but does not make this implementation competitive. These are algorithm costs, **not a lower bound on every possible representation**.

Derived: reverse cubic-functional evaluation. A birth polynomial $g(q)=\Sigma_j \beta_j q_j (a^T q)_j^2$ can be retained as a circuit. Batched query transport, evaluation of $a^T q$, its local gradient, and backward gradient transport each require a dense product. This straightforward exact architecture returns to approximately $4\times0.51=2.04$ u per source-layer. Renaming the work "automatic differentiation" does not eliminate it.

Derived: young-source arithmetic. Even deleting all 116 u of young work would leave 144 u. Adding a 20 u quartic correction leaves 164 u, with essentially no allowance to replace the young computation. To reach 150 u with that correction requires saving 130 u, more than the entire young tier. A modest young-only acceleration cannot solve the joint target.

Assessment. The measured dead list is not a universal impossibility theorem. But it supplies no evidence for a new closed, cheap state carrying the missing cubic and quartic sectors. I cannot honestly claim that turbulence, kinetic or tensor-network terminology supplies one. The leaders' estimator remains unidentified.

5. The coefficient paradox identifies a partner projection

Derived under a testable linear-response approximation. Let t=0 be the shipped K×C coefficient and t=1 restore its missing half. Write the final error vector as $e(t) \approx e_0 + t\,d$, with d the change induced by restoration. Normalize by $E_0 = \|e_0\|^2/n$:

$$\frac{E(t)}{E_0} = 1 + At + Bt^2, \quad A = \frac{2\langle e_0, d\rangle}{\|e_0\|^2}, \quad B = \frac{\|d\|^2}{\|e_0\|^2}.$$

Your reported $E(1)/E_0 \approx 1.034$ and optimum $t \approx 0.25$ imply $B \approx 0.068$ and $A \approx -0.034$. Hence

$$E(0.25)/E_0 \simeq 0.99575, \qquad \frac{\langle e_0 + d, d\rangle}{\|d\|^2} \simeq 0.75.$$

The predicted improvement is 0.425%, consistent with your −0.4…−0.8%. Thus ordinary cancellation already explains the apparent paradox. After restoring the family, the remaining error has a +0.75d projection, compensating the missing-family error −d in the shipped estimator. This does *not* identify its physical channel or prove an omitted higher cumulant.

The first experiment I would run

Measure. Record pre-activation variance, κ_3 diagonal and κ_4 diagonal trajectories for the shipped, 1.25× and fully restored runs. First test whether the 1.25× final mean equals the shipped mean plus 0.25 times the restoration response to adequate accuracy. If it does not, the quadratic explanation is insufficient.

Then replay the mean closure with each trajectory while replacing, in turn, variance, κ_3 diagonal, κ_4 diagonal, and finally all three by their teacher values. Propagate the replay's mean normally. For each intervention a compute d_a from the endpoint predictions and

$$t_a^* = -\frac{\langle e_a(0), d_a\rangle}{\|d_a\|^2}.$$

Check the predicted quadratic at t=0.25. If the κ_4 oracle shifts the optimum toward 1, it identifies κ_4 as a compensating partner *in this mean-readout intervention*; an analogous shift under the variance oracle identifies variance. Use joint interventions for cross-channel cancellation. Do not interpret unstable optima after all-three forcing leaves only teacher noise. Repeat by early/middle/late layer blocks if localization is needed.

State and FLOPs. Three trajectories of three n-vectors over 16 layers occupy under 0.6 MiB in float32; teacher vectors are already stored. One replay uses 16 matrix-vector products: $16n(2n-1)/2^{31} \approx 0.0156$ u plus O(16n) closure work. Twelve to thirty-six replays cost roughly 0.2-0.6 u plus the scalar closure arithmetic, comfortably around 1 u. Capturing missing trajectories costs the corresponding full offline chain runs; it is not a free grader operation.

End-to-end decision. Only if the κ_4 oracle exposes this cancellation would I prioritize testing restored K×C together with the mixed-coordinate quartic design. Their joint benefit is unknown. Otherwise, the local slice improvement is the wrong optimization target, and the identified partner channel should determine the next architecture. This diagnostic is more informative than another ungrounded tensor ansatz.

Sources and reproducibility

The cited sources support the source-code audit and the general derivative/connected-cumulant tools. The network-specific contractions, error budgets, cost ledgers and falsifiers are derived here. No paper is cited as a validated solution of this benchmark.

[1] 504aldo, whest-p2-cumulant-k3, estimator_v29.py. Audited the available local copy associated with revision 18c17e2d7a9aeacd399cfc2c6b571e4e16dbfb45: pre-weight shared-basis refit, newborn a_b = diag(w1) C_off, fused right-leg contractions, and the final need_d21=False branch. The current web fetch of this pinned file failed; this report does not claim a fresh GitHub-wide audit.

GitHub / estimator_v29.py / pinned source 18c17e2d

[2] Jani Lukkarinen and Matteo Marcozzi, Wick polynomials and time-evolution of cumulants, arXiv:1503.05851. General background for connected Wick/cumulant bookkeeping; it does not establish a cheap closure or benchmark accuracy here.

https://arxiv.org/abs/1503.05851

Scope of verification

A small independent NumPy check used a random n=12 tensor, not a neural-network teacher. It verified the mixed-coordinate slice identity, curvature-query identity, newborn identity cancellation, covariance-response reuse and the all-distinct subtraction. Maximum absolute discrepancies were $1.1\text{-}2.2\times10^{-16}$ in float64. This validates those algebraic identities only. It does not validate the complete quartic kernel, float32 behavior, network accuracy, runtime or memory.

No Round 9 teacher arrays or complete 140 u response implementation were supplied. The terminal cost is therefore an explicit inventory estimate, and edge/pair-matching overhead remains an item to meter. The report does not assert that the 20 u target has been met.

What is new in this response

The proposed computational boundary is a mixed-coordinate response: two legs have crossed the next weight matrix and the third has not. It exposes a new-direction K_3 query after the per-source work has been summed, while a newborn identity leg cancels the additional aggregate transport in full-slice layers. This is a different split of the work, not a new statistical approximation to old sources.

What remains unsolved

The baseline source-transport cost, the unexplained cubic all-distinct fresh error, and a validated joint improvement of all three marginal inputs remain unsolved. The compensation experiment identifies where a joint redesign would have to act. Neither the exact identities nor the cited literature establish a 150 u estimator.